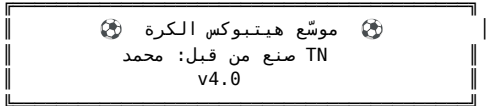


--[[



- (قابل للتغيير) RightShift : إظهار / إخفاء الواجهة
- (قابل للتغيير) H : تشغيل / إيقاف الهيتبوكس
- توقّع مسار الشوت: فيزياء كاملة (جاذبية + قوة الكرة + ارتداد + تعويض البينق)
- يدعم الكمبيوتر والجوال (زر عائِم للجوال)
- (writefile يدعم Executor إذا كان الـ) حفظ الإعدادات تلقائياً

]]

-- الخدمات

local Players = game:GetService("Players")
local UserInputService = game:GetService("UserInputService")
local RunService = game:GetService("RunService")
local TweenService = game:GetService("TweenService")
local HttpService = game:GetService("HttpService")
local Workspace = game:GetService("Workspace")

local player = Players.LocalPlayer
local playerGui = player:WaitForChild("PlayerGui")

-- الإعدادات النابتة

local CONFIG = {
 Version = "v4.0",
 Author = "محمد TN",
 SaveFile = "MohammedTN_Hitbox_v4.json",
 MinSize = 0.5,
 MaxSize = 10,
 SizePresets = { 2, 4, 6, 8 },
 AutoSizeRange = 40, -- كل 40 مسافة يتضاعف الحجم في الوضع التلقائي
 AlertSound = "rbxasset://sounds/electronicpingshort.wav",
 Prediction = {
 Step = 1 / 60, -- دقة المحاكاة (60 خطوة بالثانية)
 PointEvery = 3, -- نرسم نقطة كل 3 خطوات
 MaxBounces = 4,
 MaxBalls = 3, -- أقصى عدد كرات نتوقع مسارها بنفس الوقت
 MinSpeed = 4, -- أقل سرعة نعتبر فيها الكرة متحركة
 MaxRange = 350, -- أبعد مسافة للكرة عنك
 SettleSpeed = 2.5, -- تحت هذه السرعة العمودية الكرة تندرج ونوقف
 CalibrationRate = 0.12, -- سرعة المعايرة التلقائية
 CalibrationLimit = 80, -- نتجاهل القفزات الكبيرة (اصطدامات)
 },
 OutlineColors = {
 Color3.fromRGB(124, 92, 255),
 Color3.fromRGB(0, 190, 255),
 Color3.fromRGB(46, 204, 113),
 Color3.fromRGB(255, 200, 0),
 Color3.fromRGB(235, 77, 75),
 Color3.fromRGB(255, 255, 255),
 },
 Themes = {
 { Name = "بنفسجي", Accent = Color3.fromRGB(124, 92, 255), Accent2 = Color3.fromRGB(0, 190, 255) },
 { Name = "أحمر", Accent = Color3.fromRGB(235, 64, 90), Accent2 = Color3.fromRGB(255, 140, 60) },
 { Name = "أخضر", Accent = Color3.fromRGB(46, 204, 113), Accent2 = Color3.fromRGB(0, 200, 180) },
 { Name = "ذهبي", Accent = Color3.fromRGB(255, 190, 40), Accent2 = Color3.fromRGB(255, 120, 40) },
 },
 -- ضع روابطك هنا لتظهر في صفحة "عن السكريبت" (اتركها فارغة لإخفائها)
 Links = {
 { Name = "ديسكورد", Url = "" },
 { Name = "يوتيوب", Url = "" },
 },
}

-- الألوان

local THEME = {
 Background = Color3.fromRGB(13, 13, 20),
 Surface = Color3.fromRGB(21, 21, 32),
 SurfaceLight = Color3.fromRGB(33, 33, 50),
 Stroke = Color3.fromRGB(56, 56, 86),
 Accent = CONFIG.Themes[1].Accent,
 Accent2 = CONFIG.Themes[1].Accent2,

```

Text      = Color3.fromRGB(240, 240, 250),
SubText   = Color3.fromRGB(148, 148, 176),
Success   = Color3.fromRGB(46, 204, 113),
Danger    = Color3.fromRGB(235, 77, 75),
Warning    = Color3.fromRGB(255, 190, 40),
}

local FONT_BOLD = Enum.Font.GothamBold
local FONT_REG = Enum.Font.Gotham

-----
-- تنطيف أي نسخة سابقة من السكريبت
-----
if _G.MohammedTN_Hitbox_Cleanup then
    pcall(_G.MohammedTN_Hitbox_Cleanup)
end

-----
-- الحالة (القيم المحفوظة + قيم التشغيل)
-----
local state = {
    -- محفوظة
    ballName = "Ball",
    size = 1,
    autoSize = false,
    axisX = 1,
    axisY = 1,
    axisZ = 1,
    showHitbox = true,
    transparency = 0.5,
    outline = true,
    colorIndex = 2,
    pulse = false,
    esp = false,
    tracer = false,
    prediction = true,
    predictTime = 2.5,
    predictBounces = true,
    predictLag = true,
    predictCalibrate = true,
    predictLanding = true,
    predictClosest = true,
    predictLabels = true,
    alert = false,
    alertDistance = 15,
    toggleKey = "RightShift",
    quickKey = "H",
    themeIndex = 1,
    floatingButton = UserInputService.TouchEnabled,
    activeTab = 1,

    -- غير محفوظة
    enabled = false,
    guiVisible = true,
    minimized = false,
}

local SAVED_KEYS = {
    "ballName", "size", "autoSize", "axisX", "axisY", "axisZ", "showHitbox",
    "transparency", "outline", "colorIndex", "pulse", "esp", "tracer",
    "prediction", "predictTime", "predictBounces", "predictLag", "predictCalibrate",
    "predictLanding", "predictClosest", "predictLabels", "alert", "alertDistance", "toggleKey",
    "quickKey", "themeIndex", "floatingButton", "activeTab",
}

-----
-- حفظ / تحميل الإعدادات
-----
local canSave = typeof(writefile) == "function"
    and typeof(readfile) == "function"
    and typeof(isfile) == "function"

local function loadSettings()
    if not canSave then return end
    local ok, data = pcall(function()
        if isfile(CONFIG.SaveFile) then
            return HttpService.JSONDecode(readfile(CONFIG.SaveFile))
        end
        return nil
    end)
    if not ok or type(data) ~= "table" then return end

```

```

for _, key in ipairs(SAVED_KEYS) do
    if data[key] ~= nil and type(data[key]) == type(state[key]) then
        state[key] = data[key]
    end
end
end

-- التحقق من القيم
state.colorIndex = math.clamp(math.floor(state.colorIndex), 1, #CONFIG.OutlineColors)
state.themeIndex = math.clamp(math.floor(state.themeIndex), 1, #CONFIG.Themes)
if not pcall(function() return Enum.KeyCode[state.toggleKey] end) then state.toggleKey = "RightShift" end
if not pcall(function() return Enum.KeyCode[state.quickKey] end) then state.quickKey = "H" end
end

local savePending = false
local function scheduleSave()
    if not canSave or savePending then return end
    savePending = true
    task.delay(0.5, function()
        savePending = false
        local data = {}
        for _, key in ipairs(SAVED_KEYS) do
            data[key] = state[key]
        end
        pcall(function()
            writefile(CONFIG.SaveFile, HttpService.JSONEncode(data))
        end)
    end)
end

loadSettings()
THEME.Accent = CONFIG.Themes[state.themeIndex].Accent
THEME.Accent2 = CONFIG.Themes[state.themeIndex].Accent2

-- [part] = { Size, Transparency, CanCollide, Mass, Conn, Box, Esp, EspLabel, Tracer, Calib, LastVel, LastVelTime }
local trackedBalls = {}
local connections = {}

local function connect(signal, fn)
    local c = signal:Connect(fn)
    table.insert(connections, c)
    return c
end

-----
-- أدوات مساعدة للواجهة
-----
local accentRefs = {} -- عناصر تتغير مع الثيم
local gradients = {}

local function create(className, props, children)
    local inst = Instance.new(className)
    for k, v in pairs(props or {}) do
        if k ~= "Parent" then
            inst[k] = v
        end
    end
    for _, child in ipairs(children or {}) do
        child.Parent = inst
    end
    if props and props.Parent then
        inst.Parent = props.Parent
    end
    return inst
end

-- ربط خاصية بلون من الثيم ليتحدّث عند تغيير الثيم
local function accentize(inst, prop, key)
    inst[prop] = THEME[key]
    table.insert(accentRefs, { inst, prop, key })
    return inst
end

local function corner(radius)
    return create("UICorner", { CornerRadius = UDim.new(0, radius or 8) })
end

local function stroke(color, thickness, transparency)
    return create("UIStroke", {
        Color = color or THEME.Stroke,
        Thickness = thickness or 1,
        Transparency = transparency or 0,
        ApplyStrokeMode = Enum.ApplyStrokeMode.Border,
    })
end

```

```

    })
end

local function accentGradient(rotation)
    local g = create("UIGradient", {
        Color = ColorSequence.new(THEME.Accent, THEME.Accent2),
        Rotation = rotation or 0,
    })
    table.insert(gradients, g)
    return g
end

local function tween(obj, time, props, style)
    local t = TweenService:Create(
        obj,
        TweenInfo.new(time, style or Enum.EasingStyle.Quint, Enum.EasingDirection.Out),
        props
    )
    t:Play()
    return t
end

local function isPress(input)
    return input.UserInputType == Enum.UserInputType.MouseButton1
        or input.UserInputType == Enum.UserInputType.Touch
end

local function isMove(input)
    return input.UserInputType == Enum.UserInputType.MouseMovement
        or input.UserInputType == Enum.UserInputType.Touch
end

local function hoverButton(btn, normalColor)
    btn.MouseEnter:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = THEME.Accent, TextColor3 = THEME.Text })
    end)
    btn.MouseLeave:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = normalColor, TextColor3 = THEME.SubText })
    end)
end

-----
-- الواجهة الرئيسية
-----
local FULL_SIZE = UDim2.new(0, 360, 0, 500)
local MINI_SIZE = UDim2.new(0, 360, 0, 58)

local screenGui = create("ScreenGui", {
    Name = "MohammedTN_HitboxGui",
    ResetOnSpawn = false,
    IgnoreGuiInset = true,
    ZIndexBehavior = Enum.ZIndexBehavior.Sibling,
    Parent = playerGui,
})

-- طبقة الرسم (خطوط التتبع) خلف الواجهة
local overlay = create("Frame", {
    Name = "Overlay",
    Size = UDim2.new(1, 0, 1, 0),
    BackgroundTransparency = 1,
    Parent = screenGui,
})

local mainFrame = create("Frame", {
    Name = "Main",
    AnchorPoint = Vector2.new(0.5, 0.5),
    Position = UDim2.new(0.5, 0, 0.5, 0),
    Size = FULL_SIZE,
    BackgroundColor3 = THEME.Background,
    BorderSizePixel = 0,
    ClipsDescendants = true,
    Active = true,
    Parent = screenGui,
}, {
    corner(16),
    stroke(THEME.Stroke, 1.5),
})

local mainScale = create("UIScale", { Scale = 0, Parent = mainFrame })

create("Frame", {

```

```

        Size = UDim2.new(1, 0, 0, 3),
        BackgroundColor3 = Color3.new(1, 1, 1),
        BorderSizePixel = 0,
        ZIndex = 5,
        Parent = mainFrame,
    }, { accentGradient(0) })

```

 شريط العنوان

```

local titleBar = create("Frame", {
    Name = "TitleBar",
    Size = UDim2.new(1, 0, 0, 58),
    BackgroundColor3 = THEME.Surface,
    BorderSizePixel = 0,
    Active = true,
    Parent = mainFrame,
})

create("Frame", {
    AnchorPoint = Vector2.new(1, 0.5),
    Position = UDim2.new(1, -14, 0.5, 0),
    Size = UDim2.new(0, 36, 0, 36),
    BackgroundColor3 = Color3.new(1, 1, 1),
    Parent = titleBar,
}, {
    corner(10),
    accentGradient(45),
    create("TextLabel", {
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundTransparency = 1,
        Text = "🌐",
        TextSize = 20,
        Font = FONT_BOLD,
        TextColor3 = THEME.Text,
    }),
})

create("TextLabel", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -60, 0, 10),
    Size = UDim2.new(1, -150, 0, 22),
    BackgroundTransparency = 1,
    Text = "موسّع هينوكس الكرة",
    TextColor3 = THEME.Text,
    TextSize = 17,
    Font = FONT_BOLD,
    TextXAlignment = Enum.TextXAlignment.Right,
    Parent = titleBar,
})

create("TextLabel", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -60, 0, 32),
    Size = UDim2.new(1, -150, 0, 16),
    BackgroundTransparency = 1,
    Text = "صنع من قبل" .. CONFIG.Author .. " • " .. CONFIG.Version,
    TextColor3 = Color3.new(1, 1, 1),
    TextSize = 12,
    Font = FONT_BOLD,
    TextXAlignment = Enum.TextXAlignment.Right,
    Parent = titleBar,
}, { accentGradient(0) })

```

```

local function titleButton(text, xOffset, hoverColor)
    local btn = create("TextButton", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, xOffset, 0.5, 0),
        Size = UDim2.new(0, 30, 0, 30),
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = text,
        TextColor3 = THEME.SubText,
        TextSize = 16,
        Font = FONT_BOLD,
        Parent = titleBar,
    }, { corner(8) })
    btn.MouseEnter:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = hoverColor or THEME.Accent, TextColor3 = THEME.Text })
    end)
    btn.MouseLeave:Connect(function()

```

```

        tween(btn, 0.2, { BackgroundColor3 = THEME.SurfaceLight, TextColor3 = THEME.SubText })
    end)
    return btn
end

local closeBtn = titleButton("x", 14, THEME.Danger)
local minimizeBtn = titleButton("-", 50)

-----
-- التنويبات
-----
local TAB_NAMES = { "إعدادات ⚙️", "أدوات 🛠️", "المظهر 🎨", "التوقع 📅", "الهيتبوكس 📁" }
local TAB_COUNT = #TAB_NAMES

local tabBar = create("Frame", {
    Name = "Tabs",
    Position = UDim2.new(0, 12, 0, 66),
    Size = UDim2.new(1, -24, 0, 36),
    BackgroundColor3 = THEME.Surface,
    BorderSizePixel = 0,
    Parent = mainFrame,
}, { corner(10), stroke(THEME.Stroke, 1, 0.5) })

local tabIndicator = create("Frame", {
    Size = UDim2.new(1 / TAB_COUNT, -10, 1, -8),
    Position = UDim2.new(0, 5, 0, 4),
    BackgroundColor3 = Color3.new(1, 1, 1),
    BorderSizePixel = 0,
    Parent = tabBar,
}, { corner(8), accentGradient(0) })

local pageHolder = create("Frame", {
    Name = "Pages",
    Position = UDim2.new(0, 0, 0, 108),
    Size = UDim2.new(1, 0, 1, -108),
    BackgroundTransparency = 1,
    Parent = mainFrame,
})

local pages = {}
local currentPage

-- التنويبات الأولى على اليمين (ترتيب عربي)
local function tabPosition(index)
    return UDim2.new((TAB_COUNT - index) / TAB_COUNT, 5, 0, 4)
end

local function selectTab(index, animated)
    state.activeTab = index
    for i, page in ipairs(pages) do
        page.Frame.Visible = (i == index)
        tween(page.Button, 0.2, { TextColor3 = (i == index) and THEME.Text or THEME.SubText })
    end
    tween(tabIndicator, animated and 0.3 or 0, { Position = tabPosition(index) })
end

local function newPage(title)
    local index = #pages + 1
    local frame = create("ScrollingFrame", {
        Name = "Page" .. index,
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundTransparency = 1,
        BorderSizePixel = 0,
        CanvasSize = UDim2.new(0, 0, 0, 0),
        AutomaticCanvasSize = Enum.AutomaticSize.Y,
        ScrollBarThickness = 3,
        ScrollingDirection = Enum.ScrollingDirection.Y,
        Visible = false,
        Parent = pageHolder,
    }, {
        create("UIListLayout", {
            Padding = UDim.new(0, 10),
            SortOrder = Enum.SortOrder.LayoutOrder,
        }),
        create("UIPadding", {
            PaddingTop = UDim.new(0, 4),
            PaddingBottom = UDim.new(0, 14),
            PaddingLeft = UDim.new(0, 12),
            PaddingRight = UDim.new(0, 12),
        })
    })
end

```

```

accentize(frame, "ScrollBarImageColor3", "Accent")

local button = create("TextButton", {
    Size = UDim2.new(1 / TAB_COUNT, 0, 1, 0),
    Position = UDim2.new((TAB_COUNT - index) / TAB_COUNT, 0, 0, 0),
    BackgroundTransparency = 1,
    Text = title,
    TextColor3 = THEME.SubText,
    TextScaled = true,
    Font = FONT_BOLD,
    ZIndex = 2,
    Parent = tabBar,
}, {
    create("UITextSizeConstraint", { MaxTextSize = 12, MinTextSize = 8 }),
    create("UIPadding", { PaddingLeft = UDim.new(0, 3), PaddingRight = UDim.new(0, 3) }),
})
button.MouseButton1Click:Connect(function()
    selectTab(index, true)
    scheduleSave()
end)

pages[index] = { Frame = frame, Button = button, Order = 0 }
currentPage = pages[index]
return currentPage
end

local function nextOrder()
    currentPage.Order += 1
    return currentPage.Order
end

local function card(height)
    return create("Frame", {
        Size = UDim2.new(1, 0, 0, height),
        BackgroundColor3 = THEME.Surface,
        BorderSizePixel = 0,
        LayoutOrder = nextOrder(),
        Parent = currentPage.Frame,
    }, { corner(12), stroke(THEME.Stroke, 1, 0.45) })
end

local function label(parent, props)
    props.BackgroundTransparency = 1
    props.Font = props.Font or FONT_BOLD
    props.TextColor3 = props.TextColor3 or THEME.Text
    props.Parent = parent
    return create("TextLabel", props)
end

local function sectionHeader(text)
    local header = label(currentPage.Frame, {
        Size = UDim2.new(1, 0, 0, 20),
        Text = text,
        TextSize = 13,
        TextXAlignment = Enum.TextXAlignment.Right,
        LayoutOrder = nextOrder(),
    })
    accentize(header, "TextColor3", "Accent2")
end

local function cardHeading(parent, text, y)
    return label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, y or 10),
        Size = UDim2.new(1, -150, 0, 20),
        Text = text,
        TextSize = 15,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
end

local function cardTitle(parent, title, subtitle)
    label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, 9),
        Size = UDim2.new(1, -90, 0, 18),
        Text = title,
        TextSize = 15,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
    return label(parent, {

```

```

        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, 28),
        Size = UDim2.new(1, -90, 0, 14),
        Text = subtitle,
        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_REG,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
end

-- مفتاح تشغيل/إيقاف متحرك
local function makeSwitch(parent, initial, onChanged)
    local switch = create("TextButton", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, 14, 0.5, 0),
        Size = UDim2.new(0, 48, 0, 26),
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = "",
        Parent = parent,
    }, { corner(13) })

    local gradient = accentGradient(0)
    gradient.Parent = switch

    local knob = create("Frame", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, 3, 0.5, 0),
        Size = UDim2.new(0, 20, 0, 20),
        BackgroundColor3 = THEME.SubText,
        Parent = switch,
    }, { corner(10) })

    local value = initial
    local api = {}

    local function render(animated)
        gradient.Enabled = value
        switch.BackgroundColor3 = value and Color3.new(1, 1, 1) or THEME.SurfaceLight
        tween(knob, animated and 0.25 or 0, {
            Position = value and UDim2.new(1, -23, 0.5, 0) or UDim2.new(0, 3, 0.5, 0),
            BackgroundColor3 = value and Color3.new(1, 1, 1) or THEME.SubText,
        })
    end

    function api.set(newValue, silent)
        if newValue == value then return end
        value = newValue
        render(true)
        if not silent then
            onChanged(value)
        end
    end

    switch.MouseButton1Click:Connect(function()
        api.set(not value)
    end)

    render(false)
    return api
end

state بطاقة فيها عنوان + مفتاح مربوط بقيمة في --
local function switchCard(title, subtitle, key, onExtra)
    local c = card(54)
    local sub = cardTitle(c, title, subtitle)
    local sw = makeSwitch(c, state[key], function(value)
        state[key] = value
        scheduleSave()
        if onExtra then
            onExtra(value)
        end
    end)
    return sw, sub, c
end

-- شريط تمرير قابل لإعادة الاستخدام (الرقم يظهر فوقه على اليسار)
local activeSlider = nil

local function makeSlider(parent, y, min, max, step, initial, format, onChanged)

```

```

local page = currentPage.Frame

local bg = create("Frame", {
    Position = UDim2.new(0, 14, 0, y),
    Size = UDim2.new(1, -28, 0, 10),
    BackgroundColor3 = THEME.SurfaceLight,
    Active = true,
    Parent = parent,
}, { corner(5) })

local fill = create("Frame", {
    Size = UDim2.new(0, 0, 1, 0),
    BackgroundColor3 = Color3.new(1, 1, 1),
    BorderSizePixel = 0,
    Parent = bg,
}, { corner(5), accentGradient(0) })

local knobStroke = stroke(THEME.Accent, 2)
accentize(knobStroke, "Color", "Accent")
local knob = create("Frame", {
    AnchorPoint = Vector2.new(0.5, 0.5),
    Position = UDim2.new(0, 0, 0.5, 0),
    Size = UDim2.new(0, 18, 0, 18),
    BackgroundColor3 = Color3.new(1, 1, 1),
    ZIndex = 2,
    Parent = bg,
}, { corner(9), knobStroke })

local badge = create("TextLabel", {
    Position = UDim2.new(0, 14, 0, y - 38),
    Size = UDim2.new(0, 60, 0, 24),
    BackgroundColor3 = THEME.SurfaceLight,
    Text = "",
    TextSize = 14,
    Font = FONT_BOLD,
    Parent = parent,
}, { corner(6) })
accentize(badge, "TextColor3", "Accent2")

local api = { value = initial }

function api.set(value, animated, silent)
    value = math.clamp(math.floor(value / step + 0.5) * step, min, max)
    api.value = value
    local percent = (value - min) / (max - min)
    local t = animated and 0.25 or 0
    tween(fill, t, { Size = UDim2.new(percent, 0, 1, 0) })
    tween(knob, t, { Position = UDim2.new(percent, 0, 0.5, 0) })
    badge.Text = string.format(format, value)
    if not silent then
        onChanged(value)
    end
end

function api.update(x)
    local percent = math.clamp((x - bg.AbsolutePosition.X) / bg.AbsoluteSize.X, 0, 1)
    api.set(min + percent * (max - min), false)
end

function api.release()
    page.ScrollingEnabled = true
    tween(knob, 0.15, { Size = UDim2.new(0, 18, 0, 18) })
end

bg.InputBegan:Connect(function(input)
    if isPress(input) then
        activeSlider = api
        page.ScrollingEnabled = false
        tween(knob, 0.15, { Size = UDim2.new(0, 22, 0, 22) })
        api.update(input.Position.X)
    end
end)

api.set(initial, false, true)
return api
end

-- state بلاقة فيها عنوان + شريط مربوط بقيمة في
local function sliderCard(title, key, min, max, step, format)
    local c = card(70)
    cardHeading(c, title)

```

```

        return makeSlider(c, 46, min, max, step, state[key], format, function(value)
            state[key] = value
            scheduleSave()
        end)
    end
end

local function smallButton(parent, text, position, size)
    local btn = create("TextButton", {
        Position = position,
        Size = size,
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = text,
        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_BOLD,
        Parent = parent,
    }, { corner(6) })
    hoverButton(btn, THEME.SurfaceLight)
    return btn
end

local function horizontalRow(parent, y, height, alignment)
    return create("Frame", {
        Position = UDim2.new(0, 14, 0, y),
        Size = UDim2.new(1, -28, 0, height),
        BackgroundTransparency = 1,
        Parent = parent,
    }, {
        create("UIListLayout", {
            FillDirection = Enum.FillDirection.Horizontal,
            HorizontalAlignment = alignment or Enum.HorizontalAlignment.Left,
            Padding = UDim.new(0, 8),
            SortOrder = Enum.SortOrder.LayoutOrder,
        }),
    })
end

-- سطر معلومات (اسم على اليمين وقيمة على اليسار)
local function statRow(parent, y, name)
    label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, y),
        Size = UDim2.new(0.55, -14, 0, 18),
        Text = name,
        TextColor3 = THEME.SubText,
        TextSize = 13,
        Font = FONT_REG,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
    return label(parent, {
        Position = UDim2.new(0, 14, 0, y),
        Size = UDim2.new(0.45, -14, 0, 18),
        Text = "-",
        TextSize = 13,
        TextXAlignment = Enum.TextXAlignment.Left,
    })
end

-----
-- الإشعارات
-----

local toastHolder = create("Frame", {
    AnchorPoint = Vector2.new(1, 1),
    Position = UDim2.new(1, -16, 1, -16),
    Size = UDim2.new(0, 260, 1, -32),
    BackgroundTransparency = 1,
    ZIndex = 10,
    Parent = screenGui,
}, {
    create("UIListLayout", {
        Padding = UDim.new(0, 8),
        VerticalAlignment = Enum.VerticalAlignment.Bottom,
        HorizontalAlignment = Enum.HorizontalAlignment.Right,
        SortOrder = Enum.SortOrder.LayoutOrder,
    }),
})

local toastOrder = 0
local function notify(text, color)
    toastOrder += 1

```

```

local toast = create("Frame", {
    Size = UDim2.new(1, 0, 0, 46),
    BackgroundColor3 = THEME.Surface,
    BackgroundTransparency = 1,
    LayoutOrder = toastOrder,
    Parent = toastHolder,
}, { corner(10) })

local toastStroke = stroke(THEME.Stroke, 1, 1)
toastStroke.Parent = toast

local bar = create("Frame", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, 0, 0, 0),
    Size = UDim2.new(0, 4, 1, 0),
    BackgroundColor3 = color or THEME.Accent,
    BackgroundTransparency = 1,
    BorderSizePixel = 0,
    Parent = toast,
}, { corner(2) })

local toastLabel = label(toast, {
    Position = UDim2.new(0, 12, 0, 0),
    Size = UDim2.new(1, -28, 1, 0),
    Text = text,
    TextTransparency = 1,
    TextSize = 13,
    TextWrapped = true,
    TextXAlignment = Enum.TextXAlignment.Right,
})

tween(toast, 0.3, { BackgroundTransparency = 0.05 })
tween(toastStroke, 0.3, { Transparency = 0.3 })
tween(bar, 0.3, { BackgroundTransparency = 0 })
tween(toastLabel, 0.3, { TextTransparency = 0 })

task.delay(2.5, function()
    if not toast.Parent then return end
    tween(toast, 0.3, { BackgroundTransparency = 1 })
    tween(toastStroke, 0.3, { Transparency = 1 })
    tween(bar, 0.3, { BackgroundTransparency = 1 })
    tween(toastLabel, 0.3, { TextTransparency = 1 }).Completed:Wait()
    toast:Destroy()
end)
end

local alertSound = create("Sound", {
    SoundId = CONFIG.AlertSound,
    Volume = 1,
    Parent = screenGui,
})

-----
-- منطق الكرة
-----

local ballCountLabel, nearestLabel -- تُنشأ لاحقاً في بطاقة المعلومات

local function outlineColor()
    return CONFIG.OutlineColors[state.colorIndex]
end

local function countBalls()
    local n = 0
    for _ in pairs(trackedBalls) do
        n += 1
    end
    return n
end

local function refreshBallCount()
    if ballCountLabel then
        ballCountLabel.Text = tostring(countBalls())
    end
end

local function restoreBall(part, props)
    if part.Parent then
        part.Size = props.Size
        part.Transparency = props.Transparency
        part.CanCollide = props.CanCollide
    end
end

```

```

end

local function restoreAll()
    for part, props in pairs(trackedBalls) do
        restoreBall(part, props)
    end
end

local function untrackBall(part)
    local props = trackedBalls[part]
    if not props then return end
    restoreBall(part, props)
    props.Conn:Disconnect()
    props.Box:Destroy()
    props.Esp:Destroy()
    props.Tracer:Destroy()
    trackedBalls[part] = nil
    refreshBallCount()
end

local function addBall(part)
    if not part:IsA("BasePart") or trackedBalls[part] then return end

    local props = {
        Size = part.Size,
        Transparency = part.Transparency,
        CanCollide = part.CanCollide,
        -- الكتلة الأصلية (قبل تكبير الهيتيوكس) لحساب قوة الكرة بدقة
        Mass = math.max(part.AssemblyMass, 0.01),
        Calib = Vector3.zero,
        LastVel = part.AssemblyLinearVelocity,
        LastVelTime = os.clock(),
    }

    -- إطار يوضّح حدود الهيتيوكس
    props.Box = create("SelectionBox", {
        Adornee = part,
        Color3 = outlineColor(),
        LineThickness = 0.05,
        SurfaceTransparency = 1,
        Visible = false,
        Parent = playerGui,
    })

    -- مؤشر فوق الكرة: المسافة + السرعة
    props.Esp = create("BillboardGui", {
        Adornee = part,
        AlwaysOnTop = true,
        Size = UDim2.new(0, 150, 0, 26),
        StudsOffset = Vector3.new(0, 3, 0),
        Enabled = false,
        Parent = playerGui,
    })
    local espBg = create("Frame", {
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundColor3 = THEME.Background,
        BackgroundTransparency = 0.25,
        Parent = props.Esp,
    }, { corner(8) })
    local espStroke = stroke(THEME.Accent, 1.5)
    espStroke.Parent = espBg
    accentize(espStroke, "Color", "Accent")
    props.EspLabel = label(props.Esp, {
        Size = UDim2.new(1, 0, 1, 0),
        Text = "⚽",
        TextSize = 13,
        ZIndex = 2,
    })

    -- خط التتبع
    props.Tracer = create("Frame", {
        AnchorPoint = Vector2.new(0.5, 0.5),
        BackgroundColor3 = outlineColor(),
        BorderSizePixel = 0,
        Visible = false,
        Parent = overlay,
    })

    props.Conn = part.AncestryChanged:Connect(function(_, parent)
        if parent == nil then
            untrackBall(part)
        end
    end)
end

```

```

        end
    end)

    trackedBalls[part] = props
    refreshBallCount()
end

local function scanForBalls()
    for _, descendant in ipairs(Workspace.GetDescendants()) do
        if descendant.Name == state.ballName then
            addBall(descendant)
        end
    end
end

local function rescan()
    for part in pairs(trackedBalls) do
        untrackBall(part)
    end
    scanForBalls()
    refreshBallCount()
end

connect(Workspace.DescendantAdded, function(descendant)
    if descendant.Name == state.ballName then
        addBall(descendant)
    end
end)

local function getRootPosition()
    local character = player.Character
    local root = character and character:FindFirstChild("HumanoidRootPart")
    return root and root.Position
end

local function drawLine(frame, a, b)
    local delta = b - a
    frame.Size = UDim2.new(0, delta.Magnitude, 0, 2)
    frame.Position = UDim2.new(0, (a.X + b.X) / 2, 0, (a.Y + b.Y) / 2)
    frame.Rotation = math.deg(math.atan2(delta.Y, delta.X))
end

-----
-- توقُّع مسار الشوت (محاكاة فيزيائية) --
-----

local vizFolder = create("Folder", {
    Name = "MohammedTN_Prediction",
    Parent = Workspace.CurrentCamera,
})

local rayParams = RaycastParams.new()
rayParams.FilterType = Enum.RaycastFilterType.Exclude
rayParams.IgnoreWater = true

-- تجاهل الكرات واللاعبين ورسومنا عند فحص الاصطدام
local function refreshRayFilter()
    local list = { vizFolder }
    for part in pairs(trackedBalls) do
        table.insert(list, part)
    end
    for _, name in ipairs({ "Footballs", "Characters" }) do
        local folder = Workspace:FindFirstChild(name)
        if folder then
            table.insert(list, folder)
        end
    end
    for _, plr in ipairs(Players:GetPlayers()) do
        if plr.Character then
            table.insert(list, plr.Character)
        end
    end
    rayParams.FilterDescendantsInstances = list
end

-- مقسومة على كتلتها الأصلية (BodyForce / VectorForce) قوة الكرة
local function ballForceAcceleration(part, props)
    local total = Vector3.zero
    for _, child in ipairs(part:GetChildren()) do
        if child:IsA("BodyForce") then
            total += child.Force
        elseif child:IsA("VectorForce") and child.Enabled then

```

```

        local force = child.Force
        if child.RelativeTo == Enum.ActuatorRelativeTo.Attachment0 and child.Attachment0 then
            force = child.Attachment0.WorldCFrame:VectorToWorldSpace(force)
        end
        total += force
    end
end
return total / props.Mass
end

local function modelAcceleration(part, props)
    return Vector3.new(0, -Workspace.Gravity, 0) + ballForceAcceleration(part, props)
end

```

```

-- معايرة تلقائية: نقارن التسارع الحقيقي بالمحسوب ونصح الفرق --
local function calibrate(part, props, now)
    local velocity = part.AssemblyLinearVelocity
    if velocity == props.LastVel then return end -- ما وصل تحديث جديد من السيرفر --
    local dt = now - props.LastVelTime
    if dt > 0.005 and dt < 0.25 then
        local measured = (velocity - props.LastVel) / dt
        local residual = measured - modelAcceleration(part, props)
        local P = CONFIG.Prediction
        if residual.Magnitude < P.CalibrationLimit then
            props.Calib = props.Calib:Lerp(residual, P.CalibrationRate)
        else
            -- اصطدام أو ركلة: نخفف المعايرة بدل ما نخربها
            props.Calib *= 0.5
        end
    end
    props.LastVel = velocity
    props.LastVelTime = now
end

```

```

local function networkLag()
    local ok, ping = pcall(function()
        return player:GetNetworkPing()
    end)
    return ok and math.clamp(ping, 0, 0.5) or 0
end

```

```

-- يحاكي حركة الكرة ويرجع نقاط المسار والأحداث المهمة --
local function simulate(part, props, rootPos)
    local P = CONFIG.Prediction
    local step = P.Step
    local radius = math.min(props.Size.X, props.Size.Y, props.Size.Z) / 2
    local accel = modelAcceleration(part, props)
    if state.predictCalibrate then
        accel += props.Calib
    end

    local pos = part.Position
    local vel = part.AssemblyLinearVelocity

    -- تعويض البيق: الكرة الحقيقية متقدمة عن اللي نشوفه
    if state.predictLag then
        local lag = networkLag()
        pos += vel * lag + 0.5 * accel * lag * lag
        vel += accel * lag
    end

    -- الكرة على الأرض وتتدحرج: نلغي الجاذبية باتجاه الأرض ونرفعها شوي
    -- (فحص الاصطدام يتجاهل السطح اللي الكرة لامسته من البداية)
    local ground = Workspace.Raycast(pos, Vector3.new(0, -(radius + 0.35), 0), rayParams)
    if ground and ground.Normal.Y > 0.6 and vel:Dot(ground.Normal) < P.SettleSpeed then
        local normal = ground.Normal
        accel -= normal * accel:Dot(normal)
        vel -= normal * math.min(vel:Dot(normal), 0)
        pos = ground.Position + normal * (radius + 0.05)
    end
end

```

```

local physical = part.CurrentPhysicalProperties
local elasticity = physical and physical.Elasticity or 0.5
local friction = physical and physical.Friction or 0.3

```

```

local result = {
    Points = { pos },
    Bounces = {},
    Landing = nil, -- { Position, Time, Normal }
    Closest = nil, -- { Position, Time, Distance }
    Start = pos,
}

```

```

}

local t = 0
local stepIndex = 0
local bounces = 0

local function checkClosest(point, time)
    if not rootPos then return end
    local distance = (point - rootPos).Magnitude
    if not result.Closest or distance < result.Closest.Distance then
        result.Closest = { Position = point, Time = time, Distance = distance }
    end
end

checkClosest(pos, 0)

while t < state.predictTime do
    local nextVel = vel + accel * step
    local move = (vel + nextVel) * 0.5 * step
    local hit = move.Magnitude > 1e-4 and Workspace.Spherecast(pos, radius, move, rayParams)

    if hit then
        local fraction = math.clamp(hit.Distance / move.Magnitude, 0, 1)
        pos += move.Unit * hit.Distance
        vel += accel * step * fraction
        t += step * fraction
        table.insert(result.Points, pos)
        checkClosest(pos, t)

        local normal = hit.Normal
        if not result.Landing and normal.Y > 0.6 then
            result.Landing = { Position = pos - normal * radius, Time = t, Normal = normal }
        end
        table.insert(result.Bounces, pos - normal * radius)

        if not state.predictBounces or bounces >= P.MaxBounces then
            break
        end

        -- ارتداد: نعكس السرعة العمودية ونخفف الأفقية بالاحتكاك
        local vn = vel:Dot(normal)
        if vn < 0 then
            local normalVel = normal * vn
            local tangentVel = vel - normalVel
            vel = tangentVel * (1 - math.clamp(friction, 0, 1) * 0.25) - normalVel * elasticity
        end
        if normal.Y > 0.6 and math.abs(vel:Dot(normal)) < P.SettleSpeed then
            break -- الكرة صارت تتدحرج
        end
        pos += normal * 0.02
        bounces += 1
    else
        pos += move
        vel = nextVel
        t += step
        stepIndex += 1
        if stepIndex % P.PointEvery == 0 then
            table.insert(result.Points, pos)
        end
        checkClosest(pos, t)
    end
end

table.insert(result.Points, pos)
return result
end

-- مخزن قطع الرسم (نعيد استخدامها بدل ما ننشئ قطع كل فريم)
local pool = { Segment = {}, Bounce = {}, Landing = {}, Closest = {} }
local used = { Segment = 0, Bounce = 0, Landing = 0, Closest = 0 }

local function vizPart(shape, size)
    return create("Part", {
        Anchored = true,
        CanCollide = false,
        CanQuery = false,
        CanTouch = false,
        CastShadow = false,
        Locked = true,
        Material = Enum.Material.Neon,
        Shape = shape,
    })
end

```

```

        Size = size,
        Transparency = 1,
        Parent = vizFolder,
    })
end

local function vizLabel(adornee, offset)
    local billboard = create("BillboardGui", {
        Adornee = adornee,
        -- نرسمها فوق كل شي عشان ما تغطي الكرة --
        AlwaysOnTop = false,
        LightInfluence = 0,
        MaxDistance = 220,
        Size = UDim2.new(0, 96, 0, 20),
        StudsOffset = offset,
        Enabled = false,
        Parent = vizFolder,
    })
    create("Frame", {
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundColor3 = THEME.Background,
        BackgroundTransparency = 0.35,
        Parent = billboard,
    }, { corner(6) })
    local text = label(billboard, {
        Size = UDim2.new(1, 0, 1, 0),
        Text = "",
        TextSize = 11,
        ZIndex = 2,
    })
    return billboard, text
end

local factories = {
    Segment = function()
        return { Part = vizPart(Enum.PartType.Block, Vector3.new(0.18, 0.18, 1)) }
    end,
    Bounce = function()
        return { Part = vizPart(Enum.PartType.Ball, Vector3.new(0.6, 0.6, 0.6)) }
    end,
    Landing = function()
        local part = vizPart(Enum.PartType.Cylinder, Vector3.new(0.15, 4, 4))
        local billboard, text = vizLabel(part, Vector3.new(0, 2.5, 0))
        return { Part = part, Billboard = billboard, Text = text }
    end,
    Closest = function()
        local part = vizPart(Enum.PartType.Ball, Vector3.new(1.1, 1.1, 1.1))
        local billboard, text = vizLabel(part, Vector3.new(0, 2, 0))
        return { Part = part, Billboard = billboard, Text = text }
    end,
}

local function take(kind)
    used[kind] += 1
    local item = pool[kind][used[kind]]
    if not item then
        item = factories[kind]()
        pool[kind][used[kind]] = item
    end
    return item
end

local function hideUnused()
    for kind, list in pairs(pool) do
        for i = used[kind] + 1, #list do
            local item = list[i]
            item.Part.Transparency = 1
            if item.Billboard then
                item.Billboard.Enabled = false
            end
        end
        used[kind] = 0
    end
end

local function drawPrediction(result, color, showLabels, ballPos, clearRadius)
    local points = result.Points
    local count = #points
    for i = 1, count - 1 do
        local a, b = points[i], points[i + 1]
        local length = (b - a).Magnitude
    end
end

```

```

    -- ما نرسم داخل الكرة نفسها عشان تيان واضحة --
    local insideBall = (a - ballPos).Magnitude < clearRadius and (b - ballPos).Magnitude < clearRadius
    if length > 0.05 and not insideBall then
        local seg = take("Segment").Part
        seg.Size = Vector3.new(0.18, 0.18, length)
        seg.CFrame = CFrame.lookAt((a + b) / 2, b)
        seg.Color = color
        -- يبهت الخط كل ما بعد في المستقبل --
        seg.Transparency = 0.1 + 0.6 * (i / count)
    end
end

if state.predictBounces then
    for _, point in ipairs(result.Bounces) do
        local bounce = take("Bounce").Part
        bounce.CFrame = CFrame.new(point)
        bounce.Color = color
        bounce.Transparency = 0.25
    end
end

if state.predictLanding and result.Landing then
    local marker = take("Landing")
    local landing = result.Landing
    marker.Part.CFrame = CFrame.new(landing.Position + landing.Normal * 0.08) * CFrame.Angles(0, 0, math.pi / 2)
    marker.Part.Color = color
    marker.Part.Transparency = 0.35
    marker.Text.Text = string.format("ت ٢.٠ %", landing.Time)
    marker.Text.TextColor3 = color
    marker.Billboard.Enabled = showLabels
end

local closest = result.Closest
if state.predictClosest and closest and closest.Time > 0.05
    and (closest.Position - ballPos).Magnitude > clearRadius
then
    local marker = take("Closest")
    marker.Part.CFrame = CFrame.new(closest.Position)
    marker.Part.Color = THEME.Warning
    marker.Part.Transparency = 0.3
    marker.Text.Text = string.format("ت ٢.٠ %", closest.Distance, closest.Time)
    marker.Text.TextColor3 = THEME.Warning
    marker.Billboard.Enabled = showLabels
end
end

-----
-- الحلقة الرئيسية --
-----
local predictionLabels = {} -- تُنشأ في صفحة التوقع
local lastInfoUpdate = 0
local lastFilterUpdate = 0
local alertActive = false

connect(RunService.RenderStepped, function()
    local camera = Workspace.CurrentCamera
    local rootPos = getRootPosition()
    local nearest = math.huge
    local now = os.clock()
    local pulse = (math.sin(now * 5) + 1) / 2
    local P = CONFIG.Prediction

    if vizFolder.Parent ~= camera then
        vizFolder.Parent = camera
    end

    if now - lastFilterUpdate > 0.5 then
        lastFilterUpdate = now
        refreshRayFilter()
    end

    local moving = {}

    for part, props in pairs(trackedBalls) do
        if not part.Parent then
            props.Tracer.Visible = false
            continue
        end

        local distance = rootPos and (part.Position - rootPos).Magnitude
        if distance then
            nearest = math.min(nearest, distance)
        end
    end
end)

```

```

end

-- تطبيق الهيتبوكس
if state.enabled then
    local mult = state.size
    if state.autoSize and distance then
        mult = math.min(mult * (1 + distance / CONFIG.AutoSizeRange), CONFIG.MaxSize * 1.5)
    end
    local targetSize = props.Size * mult * Vector3.new(state.axisX, state.axisY, state.axisZ)
    local targetTransparency = state.showHitbox and state.transparency or props.Transparency

    if part.Size ~= targetSize then
        part.Size = targetSize
    end
    if part.Transparency ~= targetTransparency then
        part.Transparency = targetTransparency
    end
    if part.CanCollide then
        part.CanCollide = false
    end
end

-- الإطار + النيص
props.Box.Visible = state.enabled and state.outline
if props.Box.Visible then
    if state.pulse then
        props.Box.LineThickness = 0.03 + 0.06 * pulse
        props.Box.Transparency = 0.5 * (1 - pulse)
    else
        props.Box.LineThickness = 0.05
        props.Box.Transparency = 0
    end
end

-- المؤشر: المسافة والسرعة
local velocity = part.AssemblyLinearVelocity
props.Esp.Enabled = state.esp
if state.esp then
    -- المؤشر دائماً فوق الكرة حتى لو الهيتبوكس كبير
    props.Esp.StudsOffset = Vector3.new(0, part.Size.Y / 2 + 1.5, 0)
    props.EspLabel.Text = string.format(
        "🏏 %s 0.% < • f",
        distance and string.format("%.1f", distance) or "-",
        velocity.Magnitude
    )
end

-- خط التتبع
local ballScreen = camera.WorldToViewportPoint(part.Position)
local rootScreen = rootPos and camera.WorldToViewportPoint(rootPos)
if state.tracer and rootScreen and ballScreen.Z > 0 and rootScreen.Z > 0 then
    props.Tracer.Visible = true
    drawLine(props.Tracer, Vector2.new(rootScreen.X, rootScreen.Y), Vector2.new(ballScreen.X, ballScreen.Y))
else
    props.Tracer.Visible = false
end

-- المعايرة تشتغل دائماً عشان تكون جاهزة لما تنطلق الكرة
calibrate(part, props, now)

if state.prediction and velocity.Magnitude > P.MinSpeed and (not distance or distance < P.MaxRange) then
    table.insert(moving, { Part = part, Props = props, Distance = distance or 0, Speed = velocity.Magnitude })
end

-- توقع المسار لأقرب الكرات المتحركة
local focus
if state.prediction then
    table.sort(moving, function(a, b)
        return a.Distance < b.Distance
    end)
    local color = outlineColor()
    for i = 1, math.min(#moving, P.MaxBalls) do
        local entry = moving[i]
        local result = simulate(entry.Part, entry.Props, rootPos)
        local part = entry.Part
        -- الكتابات يس للكرة الأقرب عشان ما تزحم الشاشة
        local showLabels = state.predictLabels and i == 1
        local clearRadius = math.max(part.Size.X, part.Size.Y, part.Size.Z) * 0.6 + 0.5
        drawPrediction(result, color, showLabels, part.Position, clearRadius)
        if i == 1 then

```

```

        focus = { Entry = entry, Result = result }
    end
end
end
hideUnused()

-- تنبيه قرب الكرة
if state.alert and nearest <= state.alertDistance then
    if not alertActive then
        alertActive = true
        pcall(function() alertSound:Play() end)
        notify("⚠️ الكرة قريبة منك!", THEME.Danger)
    end
elseif nearest > state.alertDistance + 3 then
    alertActive = false
end

-- تحديث المعلومات
if now - lastInfoUpdate > 0.15 then
    lastInfoUpdate = now
    if nearestLabel then
        nearestLabel.Text = nearest < math.huge and string.format("%.1f م", nearest) or "-"
    end
    if predictionLabels.Speed then
        if focus then
            local result = focus.Result
            predictionLabels.Speed.Text = string.format("%.1f", focus.Entry.Speed)
            predictionLabels.Landing.Text = result.Landing and string.format("%.2f ث", result.Landing.Time) or "-"
            predictionLabels.Closest.Text = result.Closest
                and string.format("%.1f م (%.2f ث)", result.Closest.Distance, result.Closest.Time)
            or "-"
            predictionLabels.Calib.Text = string.format("%.1f", focus.Entry.Props.Calib.Magnitude)
        else
            predictionLabels.Speed.Text = "لا توجد كرة متحركة"
            predictionLabels.Landing.Text = "-"
            predictionLabels.Closest.Text = "-"
            predictionLabels.Calib.Text = "-"
        end
        predictionLabels.Ping.Text = string.format("%d ms", math.floor(networkLag() * 1000 + 0.5))
    end
end
end)

-----
-- صفحة 1: الهيتبوكس
-----
newPage(TAB_NAMES[1])

local function statusSuffix()
    return " • اختصار: " .. state.quickKey
end

local enableCard = card(54)
local statusText = cardTitle(enableCard, "الحالة: متوقف", "تفعيل الهيتبوكس", statusSuffix())

local function refreshStatus()
    statusText.Text = (state.enabled and "الحالة: يعمل" or "الحالة: متوقف") .. statusSuffix()
    statusText.TextColor3 = state.enabled and THEME.Success or THEME.SubText
end

local enableSwitch = makeSwitch(enableCard, state.enabled, function(value)
    state.enabled = value
    refreshStatus()
    if value then
        notify("✓ تم تفعيل الهيتبوكس", THEME.Success)
    else
        restoreAll()
        notify("تم إيقاف الهيتبوكس", THEME.Danger)
    end
end)
end)

-- الحجم + أحجام سريعة
local sizeCard = card(118)
cardHeading(sizeCard, "حجم الهيتبوكس")

local sizeSlider = makeSlider(sizeCard, 48, CONFIG.MinSize, CONFIG.MaxSize, 0.1, state.size, "x%.1f", function(value)
    state.size = value
    scheduleSave()
end)

local resetBtn = smallButton(sizeCard, "إعادة", UDim2.new(0, 80, 0, 10), UDim2.new(0, 56, 0, 24))

```

```

resetBtn.MouseButton1Click:Connect(function()
    sizeSlider.set(1, true)
    notify("تمت إعادة الحجم إلى الافتراضي", THEME.Accent2)
end)

local presetRow = horizontalRow(sizeCard, 74, 30)
local presetCount = #CONFIG.SizePresets
for i, preset in ipairs(CONFIG.SizePresets) do
    local btn = smallButton(
        presetRow,
        "x" .. preset,
        UDim2.new(),
        UDim2.new(1 / presetCount, -8 * (presetCount - 1) / presetCount, 1, 0)
    )
    btn.LayoutOrder = i
    btn.TextSize = 13
    btn.MouseButton1Click:Connect(function()
        sizeSlider.set(preset, true)
    end)
end

switchCard("حجم تلقائي حسب المسافة", "يُكبر الهيتوكس كلما ابتعدت الكرة", "autoSize")

sectionHeader("تُكبير كل اتجاه 🏏")

local axisCard = card(196)
local axisSliders = {}
local axes = {
    { key = "axisX", name = "العرض (X)" },
    { key = "axisY", name = "الارتفاع (Y)" },
    { key = "axisZ", name = "الطول (Z)" },
}
for i, axis in ipairs(axes) do
    local y = 46 + (i - 1) * 60
    cardHeading(axisCard, axis.name, y - 36)
    axisSliders[i] = makeSlider(axisCard, y, 0.25, 3, 0.05, state[axis.key], "x%.2f", function(value)
        state[axis.key] = value
        scheduleSave()
    end)
end

local axisResetCard = card(44)
local axisResetBtn = smallButton(axisResetCard, "إعادة الاتجاهات إلى x1", UDim2.new(0, 14, 0, 8), UDim2.new(1, -28, 0, 28))
axisResetBtn.TextSize = 13
axisResetBtn.MouseButton1Click:Connect(function()
    for _, slider in ipairs(axisSliders) do
        slider.set(1, true)
    end
end)

-----
-- صفحة 2: التوقع
-----
newPage(TAB_NAMES[2])

switchCard("توقع مسار الشوت", "خط ثلاثي الأبعاد يوضح وين رح تروح الكرة", "prediction")

local liveCard = card(146)
cardHeading(liveCard, "قراءة مباشرة 📺")
predictionLabels.Speed = statRow(liveCard, 38, "سرعة الكرة")
predictionLabels.Landing = statRow(liveCard, 58, "تنزل بعد")
predictionLabels.Closest = statRow(liveCard, 78, "أقرب مرور منك")
predictionLabels.Calib = statRow(liveCard, 98, "تصحيح المعايرة")
predictionLabels.Ping = statRow(liveCard, 118, "البنق")

sliderCard("مدة التوقع (ثانية)", "predictTime", 0.5, 5, 0.25, "%.2f")

sectionHeader("الدقة 🎯")
switchCard("حساب الارتداد", "يحسب ارتداد الكرة من الأرض والقائم والعارضة", "predictBounces")
switchCard("تعويض البنق", "يقدم الكرة لمكانها الحقيقي في السيرفر", "predictLag")
switchCard("معايرة تلقائية", "يتعلم انحناء الكرة من حركتها ويصح المسار", "predictCalibrate")

sectionHeader("العلامات 📍")
switchCard("مكان النزول", "دائرة مكان نزول الكرة مع الوقت", "predictLanding")
switchCard("أقرب نقطة لك", "وين رح تمر الكرة أقرب شي منك", "predictClosest")
switchCard("إظهار الكتابات", "الوقت والمسافة فوق العلامات", "predictLabels")

-----
-- صفحة 3: المظهر
-----
newPage(TAB_NAMES[3])

```

```

switchCard("إظهار الهيتبوكس", "showHitbox")
sliderCard("شفافية الكرة", "transparency", 0, 0.95, 0.05, "%.2f")
switchCard("إطار الهيتبوكس", "outline")
switchCard("تأثير النبض", "pulse")

local colorCard = card(84)
cardHeading(colorCard, "لون الإطار والمسار")

local colorRow = horizontalRow(colorCard, 42, 30, Enum.HorizontalAlignment.Right)
local colorStrokes = {}

local function selectColor(index)
    state.colorIndex = index
    scheduleSave()
    for i, s in ipairs(colorStrokes) do
        tween(s, 0.2, { Transparency = (i == index) and 0 or 1 })
    end
    local color = outlineColor()
    for _, props in pairs(trackedBalls) do
        props.Box.Color3 = color
        props.Tracer.BackgroundColor3 = color
    end
end

for i, color in ipairs(CONFIG.OutlineColors) do
    local swatch = create("TextButton", {
        Size = UDim2.new(0, 30, 0, 30),
        BackgroundColor3 = color,
        AutoButtonColor = false,
        Text = "",
        LayoutOrder = i,
        Parent = colorRow,
    }, { corner(15) })
    local s = stroke(THEME.Text, 2, 1)
    s.Parent = swatch
    colorStrokes[i] = s
    swatch.MouseButton1Click:Connect(function()
        selectColor(i)
    end)
end

local themeCard = card(84)
cardHeading(themeCard, "نيم الواجهة")

local themeRow = horizontalRow(themeCard, 42, 30, Enum.HorizontalAlignment.Right)
local themeStrokes = {}

local function applyTheme(index)
    state.themeIndex = index
    scheduleSave()
    local theme = CONFIG.Themes[index]
    THEME.Accent = theme.Accent
    THEME.Accent2 = theme.Accent2
    local sequence = ColorSequence.new(THEME.Accent, THEME.Accent2)
    for _, g in ipairs(gradients) do
        g.Color = sequence
    end
    for _, ref in ipairs(accentRefs) do
        ref[1][ref[2]] = THEME[ref[3]]
    end
    for i, s in ipairs(themeStrokes) do
        tween(s, 0.2, { Transparency = (i == index) and 0 or 1 })
    end
end

for i, theme in ipairs(CONFIG.Themes) do
    local btn = create("TextButton", {
        Size = UDim2.new(0, 64, 0, 30),
        BackgroundColor3 = Color3.new(1, 1, 1),
        AutoButtonColor = false,
        Text = theme.Name,
        TextColor3 = Color3.new(1, 1, 1),
        TextStrokeTransparency = 0.6,
        TextSize = 12,
        Font = FONT_BOLD,
        LayoutOrder = i,
        Parent = themeRow,
    }, {
        corner(8),
        create("UIGradient", { Color = ColorSequence.new(theme.Accent, theme.Accent2) }),
    })
end

```

```

    })
    local s = stroke(THEME.Text, 2, 1)
    s.Parent = btn
    themeStrokes[i] = s
    btn.MouseButton1Click:Connect(function()
        applyTheme(i)
        notify("تم تغيير النيم إلى" .. theme.Name, THEME.Accent)
    end)
end

-----
-- أدوات :4 صفحة
-----

newPage(TAB_NAMES[4])

switchCard("مؤشر الكرة", "المسافة والسرعة فوق الكرة من خلف الجدران", "esp")
switchCard("خط التتبع", "خط من شخصيتك إلى الكرة", "tracer")
switchCard("تنبيه قرب الكرة", "صوت وإشعار لما تقترب الكرة منك", "alert")
sliderCard("مسافة التنبيه", "alertDistance", 5, 50, 1, "%d م")

local nameCard = card(84)
cardHeading(nameCard, "اسم الكرة في اللعبة")

local nameBox = create("TextBox", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -14, 0, 40),
    Size = UDim2.new(1, -100, 0, 30),
    BackgroundColor3 = THEME.SurfaceLight,
    Text = state.ballName,
    PlaceholderText = "مثال: Ball",
    PlaceholderColor3 = THEME.SubText,
    TextColor3 = THEME.Text,
    TextSize = 14,
    Font = FONT_BOLD,
    ClearTextOnFocus = false,
    Parent = nameCard,
}, { corner(6), stroke(THEME.Stroke, 1) })

local searchBtn = smallButton(nameCard, "بحث", UDim2.new(0, 14, 0, 40), UDim2.new(0, 66, 0, 30))
searchBtn.TextSize = 13

local function applyBallName()
    local name = nameBox.Text:gsub("^%s+", ""):gsub("%s+$", "")
    if name == "" then
        nameBox.Text = state.ballName
        return
    end
    state.ballName = name
    scheduleSave()
    rescan()
    local n = countBalls()
    if n > 0 then
        notify("كرة باسم " .. n .. " تم العثور على", THEME.Success)
    else
        notify("لا توجد كرة باسم" .. name, THEME.Danger)
    end
end

searchBtn.MouseButton1Click:Connect(applyBallName)
nameBox.FocusLost:Connect(function(enterPressed)
    if enterPressed then
        applyBallName()
    end
end)

-----
-- الإعدادات :5 صفحة
-----

newPage(TAB_NAMES[5])

-- تغيير أزرار الاختصار
local capturing = nil -- { key = "toggleKey", button = btn }
local keyInfoLabel -- يُنشأ في بطاقة المعلومات

local function refreshKeyInfo()
    if keyInfoLabel then
        keyInfoLabel.Text = state.toggleKey .. " / " .. state.quickKey
    end
    refreshStatus()
end
end

```

```

local function keybindCard(title, key)
    local c = card(54)
    cardTitle(c, title, "اضغط على الزر ثم اختر مفتاحاً جديداً")
    local btn = smallButton(c, state[key], UDim2.new(0, 14, 0, 12), UDim2.new(0, 92, 0, 30))
    btn.TextSize = 13
    btn.MouseButton1Click:Connect(function()
        if capturing then
            capturing.button.Text = state[capturing.key]
        end
        capturing = { key = key, button = btn }
        btn.Text = "اضغط زر..."
    end)
end

keybindCard("زر إظهار الواجهة", "toggleKey")
keybindCard("زر تشغيل الهيتوكس", "quickKey")

-- الزر العائم (للجوال)
local floatingBtn = create("TextButton", {
    AnchorPoint = Vector2.new(0.5, 0.5),
    Position = UDim2.new(0, 50, 0.5, 0),
    Size = UDim2.new(0, 52, 0, 52),
    BackgroundColor3 = Color3.new(1, 1, 1),
    AutoButtonColor = false,
    Text = "🔑",
    TextSize = 24,
    Font = FONT_BOLD,
    TextColor3 = THEME.Text,
    Visible = state.floatingButton,
    ZIndex = 20,
    Parent = screenGui,
}, { corner(26), accentGradient(45), stroke(THEME.Background, 2) })

switchCard("زر عائم", "زر صغير على الشاشة لفتح الواجهة (للجوال)", "floatingButton", function(value)
    floatingBtn.Visible = value
end)

-- حفظ الإعدادات
local saveCard = card(54)
cardTitle(
    saveCard,
    "حفظ الإعدادات",
    "لا يدعم الحفظ Executor الـ" or "✓ يتم الحفظ تلقائياً"
)
local resetSettingsBtn = smallButton(saveCard, "حذف المحفوظ", UDim2.new(0, 14, 0, 12), UDim2.new(0, 92, 0, 30))
resetSettingsBtn.MouseButton1Click:Connect(function()
    if not canSave then
        notify("Executor الحفظ غير مدعوم في هذا الـ", THEME.Danger)
        return
    end
    pcall(function()
        if typeof(delfile) == "function" then
            delfile(CONFIG.SaveFile)
        else
            writefile(CONFIG.SaveFile, "{}")
        end
    end)
    notify("تم حذف الإعدادات المحفوظة - أعد تشغيل السكريبت", THEME.Accent2)
end)

sectionHeader("📊 معلومات")

local infoCard = card(86)
ballCountLabel = statRow(infoCard, 12, "الكرات المكتشفة")
nearestLabel = statRow(infoCard, 34, "أقرب كرة")
keyInfoLabel = statRow(infoCard, 56, "الواجهة / الهيتوكس")

sectionHeader("🔒 عن السكريبت")

local links = {}
for _, link in ipairs(CONFIG.Links) do
    if link.Url ~= "" then
        table.insert(links, link)
    end
end

local aboutCard = card(96 + #links * 38)

local aboutTitle = label(aboutCard, {
    Position = UDim2.new(0, 14, 0, 12),
    Size = UDim2.new(1, -28, 0, 24),

```

```

        Text = "✦ " .. CONFIG.Author .. " ✦",
        TextColor3 = Color3.new(1, 1, 1),
        TextSize = 20,
    })
    accentGradient(0).Parent = aboutTitle

    label(aboutCard, {
        Position = UDim2.new(0, 14, 0, 40),
        Size = UDim2.new(1, -28, 0, 16),
        Text = "مَوْشَع هَيْتِيوكْس الكُرَّة • " .. CONFIG.Version,
        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_REG,
    })

    label(aboutCard, {
        Position = UDim2.new(0, 14, 0, 62),
        Size = UDim2.new(1, -28, 0, 18),
        Text = "شُكْرًا لاسْتِخْدَامِك السُّكْرِبْت",
        TextSize = 13,
    })

    for i, link in ipairs(links) do
        local y = 92 + (i - 1) * 38
        label(aboutCard, {
            AnchorPoint = Vector2.new(1, 0),
            Position = UDim2.new(1, -14, 0, y),
            Size = UDim2.new(1, -120, 0, 30),
            Text = link.Name,
            TextSize = 13,
            TextXAlignment = Enum.TextXAlignment.Right,
        })
        local copyBtn = smallButton(aboutCard, "نسخ الرابط", UDim2.new(0, 14, 0, y), UDim2.new(0, 92, 0, 30))
        copyBtn.MouseButton1Click:Connect(function()
            if typeof(setclipboard) == "function" then
                pcall(setclipboard, link.Url)
                notify("تم نسخ رابط" .. link.Name, THEME.Success)
            else
                notify(link.Url, THEME.Accent2)
            end
        end)
    end

    -----
    -- السحب والإدخال
    -----
    local windowDragging = false
    local dragStart, startPos

    local floatPressed = false
    local floatMoved = false
    local floatStart, floatStartPos

    titleBar.InputBegan:Connect(function(input)
        if isPress(input) then
            windowDragging = true
            dragStart = input.Position
            startPos = mainFrame.Position
        end
    end)

    floatingBtn.InputBegan:Connect(function(input)
        if isPress(input) then
            floatPressed = true
            floatMoved = false
            floatStart = input.Position
            floatStartPos = floatingBtn.Position
        end
    end)

    connect(UserInputService.InputChanged, function(input)
        if not isMove(input) then return end

        if activeSlider then
            activeSlider.update(input.Position.X)
        elseif windowDragging then
            local delta = input.Position - dragStart
            mainFrame.Position = UDim2.new(
                startPos.X.Scale, startPos.X.Offset + delta.X,
                startPos.Y.Scale, startPos.Y.Offset + delta.Y
            )
        end
    end)

```

```

elseif floatPressed then
    local delta = input.Position - floatStart
    if delta.Magnitude > 6 then
        floatMoved = true
    end
    if floatMoved then
        floatingBtn.Position = UDim2.new(
            floatStartPos.X.Scale, floatStartPos.X.Offset + delta.X,
            floatStartPos.Y.Scale, floatStartPos.Y.Offset + delta.Y
        )
    end
end
end)

-- إظهار / إخفاء
local function setVisible(visible)
    state.guiVisible = visible
    if visible then
        mainFrame.Visible = true
        tween(mainScale, 0.35, { Scale = 1 }, Enum.EasingStyle.Back)
    else
        tween(mainScale, 0.25, { Scale = 0 }).Completed:Connect(function()
            if not state.guiVisible then
                mainFrame.Visible = false
            end
        end)
    end
end

end

connect(UserInputService.InputEnded, function(input)
    if isPress(input) then
        if activeSlider then
            activeSlider.release()
            activeSlider = nil
        end
        if floatPressed and not floatMoved then
            setVisible(not state.guiVisible)
        end
        floatPressed = false
        windowDragging = false
    end
end)

connect(UserInputService.InputBegan, function(input, gameProcessed)
    -- التقاط زر اختصار جديد
    if capturing and input.UserInputType == Enum.UserInputType.Keyboard then
        local c = capturing
        capturing = nil
        if input.KeyCode ~= Enum.KeyCode.Escape then
            state[c.key] = input.KeyCode.Name
            scheduleSave()
            notify("تم تعيين الزر: " .. input.KeyCode.Name, THEME.Success)
        end
        c.button.Text = state[c.key]
        refreshKeyInfo()
        return
    end

    if gameProcessed then return end
    if input.KeyCode.Name == state.toggleKey then
        setVisible(not state.guiVisible)
    elseif input.KeyCode.Name == state.quickKey then
        enableSwitch.set(not state.enabled)
    end
end)

-- التصغير
minimizeBtn.MouseButton1Click:Connect(function()
    state.minimized = not state.minimized
    minimizeBtn.Text = state.minimized and "+" or "-"
    tween(mainFrame, 0.35, { Size = state.minimized and MINI_SIZE or FULL_SIZE })
end)

-- الإغلاق الكامل
local function cleanup()
    state.enabled = false
    for part in pairs(trackedBalls) do
        untrackBall(part)
    end
    for _, c in ipairs(connections) do
        c:Disconnect()
    end
end

```

```
end
table.clear(connections)
if vizFolder then
    vizFolder:Destroy()
end
if screenGui then
    screenGui:Destroy()
end
end
_G.MohammedTN_Hitbox_Cleanup = nil
end
```

```
_G.MohammedTN_Hitbox_Cleanup = cleanup
```

```
closeBtn.MouseButton1Click:Connect(function()
    tween(mainScale, 0.25, { Scale = 0 }).Completed:Wait()
    cleanup()
end)
```

```
-----
-- البدء
-----
```

```
selectColor(state.colorIndex)
applyTheme(state.themeIndex)
selectTab(math.clamp(math.floor(state.activeTab), 1, TAB_COUNT), false)
scanForBalls()
refreshBallCount()
refreshRayFilter()
refreshKeyInfo()
setVisible(true)
notify("CONFIG.Author, THEME.Accent) " صنع من قبل - " .. CONFIG.Version .. " مرحباً! تم تحميل السكريت")
```